

*?? **Did You Know?** When Renaissance Technologies—the legendary quantitative hedge fund with returns exceeding 60% annually for decades—first developed their trading algorithms, they organized their system into over 1,000 specialized functions. According to former employees, these functions ranged from basic calculations (like volatility measures) to sophisticated pattern detection. This modular approach allowed their scientists to continually refine individual components without disrupting the entire system. Today, their Medallion Fund is considered perhaps the most successful investment fund in history, and its modular function-based architecture was a key innovation.*

Function Body: Implementing Your Strategy

The function body contains the actual code that executes when the function is called. This is where you implement the logic that processes the inputs to produce the desired result:

```
1. def calculate_moving_average(prices, window):
2.     # Sum the most recent prices based on the window size
3.     sum_of_prices = sum(prices[-window:])
4.
5.     # Divide by the window size to get the average
6.     moving_average = sum_of_prices / window
7.
8.     # Return the calculated average
9.     return moving_average
10.
```

The code inside the function is indented, typically by four spaces. This indentation is how Python knows which code belongs to the function. Everything at the same indentation level is part of the function body.